

UNIVERSITY OF TECHNOLOGY, JAMAICA
SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY
PROGRAMMING II (CMP1025)

LAB TEST #1

February 17, 2026

Duration 1hour :50 Mins

Python

Scenario: Airline Ticket Reservation System

Write a program that simulates an airline ticket reservation system. A user can select a destination, choose seat class, add extra luggage, select a meal preference, and checkout.

The program must display a menu that repeats until the user completes checkout or exits.

The system must track:

- Ticket base price
- Seat class upgrade cost
- Extra luggage cost
- Meal cost
- Tax
- Airport fee
- Final total

There is also a chance the passenger wins a free flight.

Program Requirements

Main Menu

The program must repeatedly display:

1. Select Destination
2. Choose Seat Class
3. Add Extra Luggage
4. Select Meal
5. Checkout
6. Exit

Destinations		Seat Class Options	
Destination Base Price		Class	Additional Cost
New York	\$45,000	Economy	\$10
Toronto	\$35,000	Business	\$20,000
London	\$75,000	First Class	\$40,000
Miami	\$40,000	Rules:	
7.		• Must select seat class before checkout.	

Extra Luggage • First bag: Free • Additional bags: \$7,000 each • Maximum 3 extra bags User must enter number of extra bags.	Meal Selection <table> <tr> <th>Meal</th><th>Cost</th></tr> <tr> <td>Standard</td><td>\$0</td></tr> <tr> <td>Vegetarian</td><td>\$2,000</td></tr> <tr> <td>Vegan</td><td>\$2,500</td></tr> <tr> <td>Premium</td><td>\$4,000</td></tr> </table>	Meal	Cost	Standard	\$0	Vegetarian	\$2,000	Vegan	\$2,500	Premium	\$4,000
Meal	Cost										
Standard	\$0										
Vegetarian	\$2,000										
Vegan	\$2,500										
Premium	\$4,000										

Checkout Rules

Tax

- 12% of subtotal

Airport Processing Fee

- Flat fee of \$6,000

Group Discount

If total flight cost (before tax) exceeds \$100,000:

- Apply 7% discount on subtotal

Validation Before Checkout

User must:

- ✓ Select destination
- ✓ Select seat class

If not, deny checkout.

Free Flight Chance

Generate a random number between 1 and 50.

If the number is 25, the flight is FREE.

Final total becomes \$0.00.

Required Functions

Each function must perform only its intended task.

1. select_destination()

- Displays destinations
- Accepts selection
- Returns base ticket price

(15 marks)

2. choose_seat_class()

- Displays class options
- Returns class upgrade cost

(15 marks)

3. add_extra_luggage()

- Accepts number of extra bags
- Validates maximum (3)
- Calculates luggage cost
- Returns cost

(15 marks)

4. select_meal()

- Displays meal options
- Returns meal cost

(10 marks)

5. checkout(subtotal)

- Validates required selections
- Applies discount (if applicable)
- Calculates tax
- Adds airport fee
- Checks free flight chance
- Displays detailed receipt

(25 marks)

Logic & Flow – 30 Marks

Comments & Documentation – 10 Marks

*******END OF EXAM*******